

1. Demographic Overview

- Large proportion of respondents are **female (88.94%)**
- A large proportion of respondents are **graduates (73.46%)**. Possibly because people voluntarily participated in the survey. There is a possible sample bias in that the insights of current students, especially those in the first and second year, are underrepresented.
- **35.3%** of respondents are under **policy government support**, while only **3.96%** of respondents indicated that they are from a **poor household**. Different respondents may define being from a poor household differently as others, meaning using policy support numbers will be more effective for future analysis.

2. Overall GPA Distribution

I split the GPA into 5 sections, by creating a calculated field, converting the numbers 1-5 to Poor, Average, Fair, Good and Excellent, based on the Codebook. As I had to pivot the tables to create the other tables and visualisations, I created an extra row in the source Excel file called ResponseID, so that I could use COUNT DISTINCT to count the total number of entries in each GPA section. I find from the visualisation that 54.79% of respondents have a "Fair" GPA level, and the second most being 31.89% of respondents with a "Good" GPA level.

3. GPA Performance by Gender

Next, I wished to show GPA distribution by gender. This dashboard shows percentage of male and female students across the 5 GPA categories mentioned earlier. In order to create the side by side pie charts comparing the percentage levels of Male and Female respectively in each GPA level, I created a new Calculated Field called Percentage Within Gender, where I divided the Count Distinct of the Response ID by the Specified Gender. Comparing the Percentages within each Gender is more beneficial than the raw numbers as there are a lot more female respondents than male. Performance between Female and Male respondents are comparable, with the females performing slightly better.

4. Demographic VS GPA Performance

Across all 3 Demographic Types, Non Minorities/ Non Policy Supported and Students from Non Poor Households perform better, with a higher percentage of Good & Excellent Performers. However, Non Minorities/ Non Policy Supported and Students from Non Poor Households have a lower percentage of students in the "Fair" GPA category. These dashboards allow system administrators to see if less privileged students are performing worse, allowing them to carry out support programs as needed.

5. Parents Background Overview

This dashboard shows the distribution of parent's occupation and education levels of the students, stacked to allow us to compare their Mother's and Fathers. This data is useful to understand the background of the students, and can help us analyse if the parent's background has any effect on the

nurturing of their children, and hence their performance in school. It is also good to view the general demographics of the students' parents at school.

6. Parents Background Overview

Further developing the analysis from the previous dashboard, we now compare student GPA performance with parental occupation and education categories. Both tables are presented in a heatmap format, where a darker shade of blue highlights a higher percentage of Respondents in that category. By Computing the Percentage total, down the table, for each GPA level we are able to see the percentage distribution of the parents education levels. For example, under students with an Excellent GPA, 44.86% of their Parents have a University education, which suggests that Parents who have higher education are more statistically likely to support children who perform better in school.

7. Student Activity VS GPA performance

Calculate Percentages Down (Each GPA Level is 100%)

To see insights of how students in each GPA level spend their time.

Social Media:

- Majority of students in Poor category spend over 4 hours on Social Media (30.14%)
- Majority of students in Excellent category spend over 2-3 hours on Social Media (37.38%)
- Spending too much time is bad, but having balanced time still can allow for excellent results

Time Studying:

- Across all GPA levels, students mostly spend over 8 hours studying
- Students who are excellent performers spend at least 4 hours studying
- While spending time is important, efficiency of studying is important as well

Time with Friends:

- Across all GPA levels, students mostly 1-3 hours with friends
- Excellent and Good performers mostly spend 1-2 hours with friends
- Not as clear impact as Social Media and Time Studying

8. Factors Score Comparison

In the Survey, respondents answered on a scale from 1 – 5, 1 being Not at all agree to 5 being Very likely Agree on different factors affecting their Education. The graph below Compares the average score from all respondents for the factors in ascending order. From this we can see that having a quality lecturer is viewed as the most important factor while being able to Adapt to the learning environment in university is the least. The graph above shows a Gantt chart that shows the average volume of Respondents scores at different GPA levels. There is a filter on the right that allows users to toggle between the different factors.

9. Factors Scores VS GPA Performance

Finally, the last dashboard shows a heat map between the different factors and GPA labels, this time I decided to display the numbers as well as it is a rather large table with many factors. Splitting it by GPA levels allows us to understand better what high performing students view as important and vice versa, which can allow counsellors and teachers to advice poor performing students. As we can see,

students with an Excellent level GPA view having Quality Lecturers as being the most important factor in school, while Students with a Poor GPA view having an ideal Training Curriculum as the most important. Colour coding the average factor scores allows us to point out extreme values with ease.